

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation quality within Qatar is a relative strength: 6 annotators + 3 supervisors, all Arabic-language teachers/lecturers with 5 holding advanced degrees [\[Q30, Q31\]](#), with detailed guidelines, training, moderation, and adjudication procedures [\[Q41, Q42, Q43\]](#). Inter-annotator agreement is substantial, averaging QWK 0.66 (P5) to 0.75 (P1) [\[Q44\]](#). However, P5 required third-annotator adjudication on 23.3% of essays [\[Q45\]](#), indicating interpretive instability on explanatory prompts. The decisive concern is institutional homogeneity: rubric, annotators, schools, and funding are all Qatar-anchored (QUTC, QRDl, Qatar MoE) [\[Q33, Q84, Q86\]](#), with no documented non-Qatari annotators and no cross-national norm validation (gap_id 1, gap_id 6). For deployment in Egypt, Saudi Arabia, Jordan, and Lebanon, it is unknown whether STY and DEV labels reflect pan-Arab teacher consensus or Qatar-specific institutional convention [\[WEB-5, WEB-6, WEB-7\]](#). OC was rated MODERATE priority, and the in-region (Qatar) signal is genuinely positive — but the cross-national transferability is unsubstantiated. A score of 3 reflects strong within-Qatar content validity tempered by unvalidated transferability.

ID	Evidence
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q41	"To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized." (p.6)
Q42	"The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes." (p.6)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q44	"We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency." (p.6)
Q45	"However, variability emerges by prompt and trait. In particular, P5 showed the lowest average agreement, noting that it has the highest percentage of essays that require a third annotator (A3: 23.3%; Table 3). This suggests greater initial disagreement" (p.6)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q84	"We heartily thank our dedicated annotators for their contributions and express our gratitude to Qatar University Testing Center, the Ministry of Education and Higher Education in Qatar (the Arabic Section of the Department Of Educational Supervision in particular), participating schools, and students for making this work possible." (p.9)
Q86	"This work was made possible by NPRP grant NPRP14S-0402-210127 from the Qatar Research Development and Innovation (QRDI) Council." (p.9)

WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)

Strengths

- All annotators qualified Arabic-language educators, 5 with MSc/PhD [Q30, Q31]
- Substantial inter-annotator QWK agreement averaging 0.66–0.75 [Q44]
- Comprehensive guidelines with terminology, exemplars, training, moderation [Q41, Q102]
- Annotator-blind scoring with anonymization [Q42, Q98]
- Adjudication procedure for high-discrepancy essays [Q43]
- Fair compensation for annotators at or above local professional rates [Q101]

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Q102	"They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures." (p.10)
Q42	"The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes." (p.6)
Q98	"Annotation was conducted on a platform that prevented annotators from accessing student identities or demographic information." (p.10)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q101	"All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators

OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Labels reflect Qatari Arabic-language-teacher perspectives [Q30, Q31, Q33, Q84]. Whether they reflect Egyptian, Saudi, Jordanian, or Lebanese stakeholder perspectives is undocumented [gap_id 1].

OC-2: Potential disagreement is plausible for subjective traits (STY, DEV) but unmeasured. Lebanese student MSA-register avoidance [WEB-9] suggests a different teacher community may judge stylistic features differently.

OC-3: Annotator demographics partially documented: 6+3 team, Arabic-language educators, 5 with advanced degrees [Q30, Q31]; nationality and per-trait IAA breakdown not in published paper [gap_id 6].

OC-4: Re-annotation by a representative cross-national pool would be needed before deploying outside Qatar.

OC-5: Aggregation rule (mean rounded down, supervisor adjudication if HOL diff ≥ 6) [Q43] could erase minority annotator perspectives, though escalation provides a safety valve. Within-Qatar perspectives may also collapse to a single institutional norm.

OC-6: Documented label issues: single-institution annotator pool [Q84], unmeasured cross-national norm transferability, P5 elevated adjudication rate [Q45].

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WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)
WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
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Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
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Information Gaps

- Per-trait IAA breakdown (especially STY, DEV)
- Annotator nationality / non-Qatari Arab participation
- Cross-national teacher-judgment correlation studies

Requires Expert Verification

- Sample re-annotation by Egyptian, Saudi, Jordanian, Lebanese teachers to estimate cross-national QWK
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Remediation

Gap 1: All annotators are Qatar-affiliated; cross-national norm transferability for subjective traits (STY, DEV) is empirically unstudied.

Recommendation: Recruit a representative pool of Egyptian, Saudi, Jordanian, and Lebanese Arabic-language teachers to re-annotate a stratified subsample of LAILA essays; report cross-national QWK per trait to quantify norm divergence before deploying outside Qatar.